

Diabetes Predictive Modeling: Capstone Project

Introduction to the Data: The data for this project comes from a dataset on Kaggle, [Diabetes Health Indicators](#), which collects information from the CDC's annual BRFFS telephone survey. This particular dataset takes 2015 survey data, and is a subset of the much larger data pool that focuses specifically on diabetes health indicators. There are 22 features and 253,680 responses, ranging from three potential diabetic classifications: not having diabetes, prediabetes, and having diabetes.

Database: The data is being stored in a PostgreSQL database through pgAdmin 4. By utilizing this approach to database management, I can develop a schema that works with my data and query efficiently using SQL. Those queries can then easily contribute to the remainder of the project and generate insights on modeling approaches and overall comprehension of the dataset.

Problem Background: Diabetes is one of the most common diseases in the United States, and can be accompanied by heart disease, kidney disease, limb amputations, and other severe health complications. The data from this survey shows the corresponding health indicators for people that do or do not have diabetes, which is useful in itself, but it also provides the prediabetes class. This is highly significant because the prediabetic classification can allow individuals to communicate with the necessary health professionals to address their futures. [SURVEY](#)

Problem Statement: Can an individual's health indicators, such as nutritional or physical habits, accurately predict diabetes cases?

Hypothesis: High blood pressure and history of limited to no physical activity contribute greatly to diabetic cases.

Methodology: The task for this project is prediction. A tree-based model (e.g. Random Forest) and a distance-based model (e.g. SVR) will be tested. Based on the results and tuning approaches used, ensembling could be implemented to further the quality of the project.

Explanation of Code: In the first python file, 'diabetes_cleaning_dataset', the code mainly consisted of getting a baseline understanding of the data and doing some preprocessing to make sure the data was clean for further analysis. This included changing the names of columns to be lowercase and more concise in certain situations (e.g. heartdiseaseorattack became

heart_disease). There are also a few different types of scale in the survey used to collect this data, so I confirmed the scale type with a `.max()` function. Additionally, I looked at duplicates in the dataset, but I came to the conclusion, at least temporarily, that due to the amount of binary features and number of respondents, the duplicates were more likely individuals with the same information than actual duplicates.

To create the database schema, I first made a few tables that made loading raw data easier, which led to confirming additional scales used in the CDC survey to make sure any buckets created were accurate, such as with age brackets or income levels.

'diabetes_eda' is where the deeper analysis started. This mainly consisted of generating graphs to get a sense of the distribution of the data, specifically the class imbalance which will need to be addressed in future steps. Outside of the class distribution graph, the heatmap was very helpful in getting an understanding of what features are heavily correlated to the diabetes status. This file was supported by LLMs in a couple of areas. When generating the graphs, the `zip()` function allowed the different axes to be paired with the corresponding variable, which was something I used at LLM to work through. I also used it as a way to remember all the arguments for the graphs.